BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING | GROUP B – EVE

WEEK 6 PROGRESS REPORT

Project: Centenary Bank Help Desk Agent

Week ending: Week 6 (5–9 October 2026) | Memory, State and Interoperability

1. Objective and weekly summary

Week 6 focused on explicit workflow state, justified case-history memory, and interoperability. Building on the bounded agent loop and tool orchestration, the group produced a state model, a memory and data-handling note, a memory/state prototype with four scenarios, and an MCP-style interface specification. The Week 6 demo was run, and all four scenario assertions passed in the terminal output provided for this report.

The designs keep identity, authorization and consequential actions under trusted application control. The proposed MCP connection remains a design only and has not been integrated or run.

2. Work completed

- **Workflow state model.** Documented the request lifecycle from classification and retrieval through action selection, approval, tool execution, observation and a terminal outcome. It sets a maximum of four workflow iterations and one safe retry per failed call.
- **Memory design and data handling.** Selected authenticated customer case history as the one persistent-memory use case. The record is limited to a ticket reference, issue category, short summary, status, last action and update time. The note defines access, retention, deletion and minimization rules.
- **Memory/state implementation.** Added a Week 6 controller and an in-process MemoryStore shared across requests during one demo run. It loads records for the trusted session customer and writes a validated, bounded summary after confirmed ticket creation. The store is not durable across restarts, and the demo uses a deterministic planner.
- **Demonstration results.** The provided terminal output shows all four scenarios completed with passing assertions: ticket created and saved to memory (TICKET_CREATED); returning customer's ticket status retrieved (TICKET_STATUS_RETURNED); cross-customer access denied (AUTHORIZATION_DENIED); and escalation completed after a simulated approval (ESCALATED_AFTER_APPROVAL). Scenario 2 uses a scripted ticket ID, so it does not isolate memory as the reason for the lookup. The run transcript is archived at evidence/traces/week6/demo-week6-run-2026-10-09.txt.
- **MCP-style interface specification.** Specified a proposed ticket-status lookup, its data contract, security boundaries and verification plan. The design is unimplemented: no ticket database, MCP server or client exists and no MCP call has been made.

3. State, memory and control decisions

The state model separates per-request workflow information from case-history memory. Request identity comes from the authenticated session; evidence and tool calls are retained in the workflow; approval status is controlled by trusted application code or a human; and the final response and stop reason are recorded at completion. The model may propose an action, while trusted code validates and executes it.

The documented use case is continuity for a returning customer with an open case. In the demo, the controller loaded the customer's stored case and the ticket lookup returned its status without a ticket ID in the customer's message. The scripted planner supplied that ID, so a run with the memory-aware Gemini planner would give stronger evidence that memory drives the selection. Ownership checks bind reads to the session customer; memory cannot change identity, authorize escalation or determine a financial outcome. The escalation scenario also shows the approval gate still runs, using a simulated approval callback.

4. Privacy, safety and prototype limits

The memory design excludes passwords, PINs, approval tokens, full account numbers, card security codes and unrestricted conversation transcripts. The prototype rejects disallowed memory fields, requires core case fields, limits the stored summary to 200 characters and records memory read, write and delete events without placing the deleted case content in the audit event. Deletion and retention are addressed in the design note; the current in-process demonstration store does not provide production persistence or a production deletion service.

All ticket and customer records in the demonstration are synthetic. The prototype is not connected to Centenary Bank systems and does not establish production-grade authentication, encryption, database security, retention periods or regulatory compliance; these will be resolved before any real-data deployment.

5. Deliverable status and task allocation

Deliverable	**Status and repository evidence (https://github.com/MariamW20/Group_B_AI_Agentic_Capstone), (https://app.clickup.com/1200450000000085/v/li/1200450000081679)**	**Owner**
State model	Documented: docs/requirements/week6/Week6_State_Model.md	WalusimbiAshraf
Memory design and data handling	Documented: docs/requirements/week6/Week6_Memory_Design_and_Data_Handling_Note.md	WalusimbiAshraf
Memory/state implementation	Prototype code: evidence/demo/centenary-agent-week6/	Mugoya Yusuf
Demonstration evidence	Run completed; all four scenario assertions passed. Transcript: evidence/traces/week6/demo-week6-run-2026-10-09.txt. Scenario 2 uses a scripted ticket ID.	Mugoya Yusuf
Interoperability	MCP-style specification complete as a design.	Ssenoga Herman
Progress report	This document.	Wambui Mariam

6. Overall assessment

The group delivered the Week 6 designs, prototype and MCP-style specification, and the four-scenario demo ran successfully, supporting the state, memory, ownership-denial and approval-gate flows.

7. Next steps: Week 7 (12–16 October 2026)

Week 7 (Evaluation, Observability and Guardrails) will show the agent is reliable, grounded, inspectable and safe: a 30-scenario evaluation set (normal, edge, incorrect-information, adversarial, tool-failure, unauthorized-action); measurable criteria; logs and traces; validation, tool allow-list, authorization, iteration limits, approval controls and prompt-injection awareness; and a Failure Catalogue of five or more failures, re-tested after fixes.